

AILT - moderation by distinct vs shared attributes (#190879)

Author(s)

This pre-registration is currently anonymous to enable blind peer-review.
It has 3 authors.

Pre-registered on: 09/19/2024 10:17 AM (PT)

1) Have any data been collected for this study already?

No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?

We are examining the relationship between AI literacy and preferences for AI (versus humans) to complete tasks. Specifically, in other studies, we show that people with lower AI literacy have generally higher AI receptivity across a broad range of tasks. However, we predict that this receptivity is moderated by the types of attributes associated with completing the tasks. Specifically, we predict that the relationship between lower literacy and greater AI receptivity is greater for tasks associated with distinctly human attributes (as defined by Santoro and Monin, 2023), and is moderated when tasks are associated with shared attributes.

3) Describe the key dependent variable(s) specifying how they will be measured.

The dependent measure will ask participants: "For each of the following tasks, please rate the extent to which you would prefer to have the key decision for that task carried out by a human or by an AI algorithm" for 10 different tasks on a 7-point scale (1 = definitely prefer the human, 7 = definitely prefer the AI algorithm). The 10 tasks cover cooking, art, architecture, news reporting, fashion, advertising, customer experience innovation, customer engagement and retention, healthcare, and hospitality management.

4) How many and which conditions will participants be assigned to?

There will be 2 between-subjects conditions: distinct attributes vs. shared attributes. In the distinct attributes condition, the tasks are associated with more distinct attributes (as measured in a pre-test) as compared to the shared condition.

Additionally, we will be measuring participants' AI literacy as measured using a 17-item multiple choice test. The sum of the correct responses will serve as their AI knowledge level, and will be used as a separate independent variable.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

We will calculate participants' average AI receptivity across the 10-tasks. We will then regress this receptivity rating on their AI literacy score (mean-centered), condition (0 = shared, 1 = distinct), and their interaction.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

We include two attention checks at the beginning of the survey. The first will ask people what answer does not correspond to the picture they see, and the second will ask people what number is visible in a separate image. People who do not answer these correctly will be excluded from taking the survey.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

We will request 1,000 completed surveys from Toluna, a market research company that has agreed to run our survey among their panel.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

There will be a number of other measures collected that will be collected for Toluna (the company running the survey). These questions are unrelated to the current research and will be collected after the measures listed in this pre-registration. Demographic information will be provided by Toluna.